

## Michaela K. Shallue

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Utah Water Research Laboratory  
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## Education

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- PhD, Civil and Environmental Engineering**, *Utah State University* Jan 2024 – Present  
River mechanics and modeling | Advisor – Colin B. Phillips  
Dissertation: Disturbance driven sediment dynamics and channel response
- MSc, Geology**, *Utah State University* Aug 2021 – Dec 2023  
Geoscience education | Advisor – Joel Pederson  
Thesis: Place-based storm hydrology for Native Student Engagement in Geoscience  
Minor: Climate Adaptation Science
- BSc, Geology**, Honors, *Northern Arizona University* Aug 2015 – May 2019  
Fluvial geomorphology | Advisor – Taylor Joyal  
Thesis: Hydrologic impacts of climate and land use in Canyon de Chelly, Arizona  
Minors: Honors Program, Psychological Sciences

## Academic Appointments

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- Research and Teaching Assistant, *Utah State University* 2024 – Present  
Utah Water Research Laboratory, Civil and Environmental Engineering Department
- Research and Teaching Assistant, *Utah State University* 2021 – 2023  
Department of Geosciences
- Climate Adaptation Science Fellow, *Utah State University* 2021 – 2023  
Ecology Center

## Teaching Experience

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### Utah State University

Role: Graduate student | 5 courses

**Environmental and Hydrologic Data Analysis and Experimentation**, Spring 2026  
CEE5930/6660. Teaching Assistant.

Course size: 15 graduate students, 1 undergraduate.

**Fluid Mechanics**, CEE3500. Teaching Assistant. Fall 2025  
Course size: 69 undergraduates.

**Geomorphology Lab**, GEO/WATS 3600L. Instructor (2 years). Fall 2022, 2023  
Course size: 24 undergraduates.

**Geomorphology Lecture**, GEO/WATS 3600. Teaching Assistant (2 years). Fall 2022, 2023  
Course size: 24 undergraduates.

**Field Camp**: Silver Islands, UT, Project, GEO 5200/5210. Teaching Assistant. Summer 2023  
Course size: 13 undergraduates (8 Utah State, 5 Weber State).

### Guest lectures

Physical Geology (Fall 2025), GEO1110. Guest lecture on fluvial geomorphology.

Introduction to Geology (Summer 2022, 2023), GEO1010. Guest lecture on runoff processes.

## Publications

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### Peer-Reviewed (1)

1. Masteller, C.C., Phillips, C.B., Kostynick, R.P., Castejon-Villalobos, J.F., Lopez, C.G., **Shallue, M.K.**, Bower, J., Sigman, A., (2026). Tracking the trajectory of alluvial channel adjustment reveals along-river shifts in sediment mobility. *Geophysical Research Letters*, 53, doi.org/10.1029/2026GL122483

### Non-Peer-Reviewed (2)

1. Pederson, J., Tanski, N., **Shallue, M.**, 2024. Tracking the timing and dynamics of Colorado River incision through study of river terraces near Dewey, UT, *Final Report on Canyonlands Natural History Association Discovery Pool Grant*.
2. **Shallue, M.**, 2023. Building and Using a Hydrology Experiment for Place-Based Learning with Native American Students, *Utah State University MSc Thesis*.

### Other (1)

1. **Shallue, M.**, 2021. Earth Notes: The Returning Rapids of Lower Cataract Canyon, *Earth Notes: KNAU News and Talk - Arizona Public Radio*.

## Conference Presentations

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Presentations – Total: 12\* | Oral: 7 | Poster: 6

\*One presentation combined both formats and is counted in both categories

- 2026** (2)      **Shallue, M.**, Patterson, N., Scamardo, J., Phillips, C.B., 2026. Rethinking Flood Frequency in Dryland Streams: Hydrologic and Geomorphic Drivers of Flood Hazard. 2026 Fall Meeting, AGU, *oral presentation*. (in prep)
- Shallue, M.**, Patterson, N., Scamardo, J., Phillips, C.B., 2026. Hydrologic and Geomorphic Controls on Flooding in Dryland Riverscapes. Geological Society of America Connects 2026, *poster*. (in prep)
- 2025** (4)      **Shallue, M.**, Scamardo, J., Phillips, C.B., 2025. Beyond the Annual Maximum Series: Hydrological and Geomorphic Flood Risk in Dryland Streams. 2025 Fall Meeting, AGU, *poster*.
- Shallue, M.**, Phillips, C.B., 2025. Patterns in Suspended Sediment Reveal Dynamic River Transport Response to Dam Removal. 2025 Fall Meeting, AGU, *GeoBurst (oral + poster presentations)\**.
- Shallue, M.**, Pederson, J., Searle, K., 2025. Cultivating Interest in Geoscience Among Native Students with an Outdoor Hydrology Experiment. Geological Society of America 2025 Rocky Mountain Section Meeting, *oral presentation*.
- Shallue, M.**, Phillips, C.B., 2025. Dam Removal Reveals Influence of Sediment Availability on Transport Dynamics. 2025 USU Spring Runoff Conference, *poster*.
- 2024** (2)      **Shallue, M.**, Phillips, C.B., 2024. Dam Removal Reveals Influence of Sediment Availability on Transport Dynamics. 2024 Fall Meeting, AGU, *poster*.
- Shallue, M.**, 2024. Dam Removal Influence on Sediment Transport. TAPDINTO-STEM Mountain Hub Convening, *oral presentation*.

- 2023** (1) **Shallue, M.**, Chavez, E., 2023. Evapotranspiration and Water Trends in the San Luis Valley. State of the Basin 2023, *oral presentation*.
- 2022** (1) **Shallue, M.**, Huntley, N., 2022. Interdisciplinary Specialization for Graduate Students Who Want to Put Science into Action: Climate Adaptation Science (CAS). Geological Society of America Connects 2022, *oral presentation*.
- 2019** (2) **Shallue, M.**, Joyal, T.J., Shaw, N., 2019. Canyon de Chelly: Watershed Modeling to Predict the Hydrological Impacts of Climate and Land Use. Student Water Symposium, Northern Arizona University, *oral presentation*.
- Shallue, M.**, Joyal, T.J., Shaw, N., 2019. Canyon de Chelly: Watershed Modeling to Predict the Hydrological Impacts of Climate and Land Use. Undergraduate Research Symposium, Northern Arizona University, *poster*.

## Grants

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Awarded (4) - To date: \$7,000 awarded as lead PI.

Utah State University | Role: Graduate Student | 3 grants (3 PI)

**Geological Society of America, GSA**, Graduate Research Student Grant 2026  
[Student, \$3,000]

**Geological Society of America, GSA**, Graduate Research Student Grant 2022  
[Student, \$2,500]

**Four Corners Geological Society**, Master's Degree Thesis Grant 2022  
[Student, \$1,000]

Northern Arizona University | Role: Undergraduate Student | 1 grant (1 PI)

**School of Earth and Sustainability**, Virginia L. Korte Undergraduate Research Award 2018  
[Student, \$500]

Student Travel Grants (4) - To date: \$1,250 awarded (2024 - 2026)

Awarded from: USU School of Graduate Studies, USU College of Engineering, and the Rocky Mountain Section of the Geological Society of America, to attend the AGU Fall Meeting or GSA Connects.

## Honors & Awards

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Farouk El-Baz Student Research Award, Geological Society of America 2026

TAPDINTO-STEM Mentorship, TAPDINTO-STEM Alliance 2023 - 2024, 2024 - 2025

Outstanding Graduate Teaching Assistant, USU Department of Geosciences 2024

J. Stewart Williams Graduate Fellowship, USU Department of Geosciences 2022, 2023

Climate Adaptation Science Fellowship, Utah State University 2021 - 2022

Outstanding Field Camp Student, NAU Geology Department 2019

1st Place Undergraduate Presentation, NAU Student Water Symposium 2019

Mary Crawley Scholarship, NAU Cline Library 2019

Lumberjack Scholars Award, Northern Arizona University 2015 - 2019

## Leadership & Service

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### Departmental Service

|                                                                              |                           |
|------------------------------------------------------------------------------|---------------------------|
| Civil and Environmental Engineering Department, <i>Utah State University</i> |                           |
| Lead Student Ambassador, Utah Water Research Laboratory                      | Fall 2025 - Summer 2026   |
| Student Research Symposia Judge, USU Office of Research                      | April 2026                |
| Student Ambassador, Utah Water Research Laboratory                           | Fall 2024 - Summer 2025   |
| Department of Geosciences, <i>Utah State University</i>                      |                           |
| Rock and Fossil Day Volunteer                                                | Feb 2024, 2025            |
| Project Co-Lead, Native American Summer Mentorship Program                   | Summer 2022, 2023         |
| URGE Pod Member, USU URGE Refinement Committee                               | Fall 2022 - Summer 2023   |
| Committee Member, Ecology Center Seminar Series                              | Spring 2022 - Spring 2023 |
| DEI Graduate Representative, Department of Geosciences                       | Fall 2021 - Fall 2023     |
| Co-Lead, Geosciences DEI Brown Bag Lunches                                   | Spring 2023 - Fall 2023   |
| School of Earth and Sustainability, <i>Northern Arizona University</i>       |                           |
| Research Volunteer (post-bacc)                                               | Sept 2019 - May 2020      |
| Undergraduate Student Researcher                                             | Aug 2018 - May 2019       |
| Cline Library, <i>Northern Arizona University</i>                            |                           |
| Department Representative, Library Spaces Committee                          | Oct 2019 - June 2021      |
| Student Ambassador                                                           | Aug 2018 - May 2019       |

### Scientific Meetings

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|--------------------------------------------|------------|
| Geological Society of America              |            |
| Student Volunteer, Annual Connects Meeting | 2022       |
| Student Volunteer, Joint Section Meeting   | 2018, 2022 |

### Training

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|-------------------------------------------------------------------------------|--------------------|
| CLaSH Summer Field School (Graduate Technical Short Course), <i>CLaSH</i>     | July 2026          |
| EPSP Graduate Student Mentorship Program (mentee), <i>AGU EPSP Section</i>    | Fall 2025          |
| TEACH Cultural Competence Training, <i>Utah State University</i>              | Jan 2025           |
| Empowering Teaching Excellence (ETE), <i>Utah State University</i>            |                    |
| ETE Explore College Teaching Program                                          | Jan 2024 - Present |
| ETE Learning circles:                                                         |                    |
| Anti-Racist Community Engagement Principles and Practices                     | Sept 2026          |
| The Norton Guide to Equity-Minded Teaching                                    | Feb 2025           |
| "Whatever It Is, I'm Against It": Resistance to Change in Higher Education    | Oct 2024           |
| TAPDINTO-STEM Mentorship Program (mentee), <i>USU, NSF</i>                    | 2023 - 2025        |
| Science, Management, & Policy Exchange, <i>USU Climate Adaptation Science</i> | March 2022, 2023   |
| Glacial Geomorphology and Volcanology Field Camp, <i>South Dakota Mines</i>   | Jul/Aug 2018       |

### Other Service and Roles

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|--------------------------------------------------|----------------------|
| Prehistoric Museum, <i>Utah State University</i> |                      |
| Education and Paleontology Intern                | July 2022 - Aug 2022 |
| Field and Preparatory Lab Volunteer              | Aug 2021 - Present   |
| Science Moab                                     |                      |
| Media Outreach Volunteer                         | Oct 2020 - Aug 2021  |
| Sierra Club, Grand Canyon Chapter                |                      |
| Student Volunteer                                | Oct 2020 - Sept 2021 |
| Hopi Water Resources                             |                      |
| Water Data Manager                               | Oct 2018 - May 2019  |

Cline Library, Northern Arizona University  
E-Resources Library Coordinator  
Assembly Hall Student Assistant  
Course Reserves Student Assistant  
User Experience Student Assistant

Dec 2018 – June 2021  
Aug 2018 – Dec 2018  
May 2018 – Dec 2018  
Nov 2015 – Dec 2018

## Skills

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### Field Experience

Techniques: River surveying; modal clast counting, Wolman pebble counting; intercept vegetation surveys; differential GPS, stadia rod surveying; bedrock and surficial mapping; sampling for optically stimulated luminescence (OSL) dating; paleontological prospecting, quarrying, and field jacketing.

Locations: Canyon de Chelly, Arizona; Book Cliffs, Utah; South Fork Eel River, California; Wasatch Plateau, Utah; Big Water, Utah; Colorado River near Dewey Bridge and Glen Canyon, Utah; Bear River Mountains, Utah; Rocky Mountains near Fort Collins, Colorado.

### Laboratory Experience

Techniques: OSL sample preparation; wet and dry sieving; vertebrate fossil preparation, archival vertebrate fossil jacketing.

### Software Experience

Adobe Creative Suite (Illustrator, Photoshop, and InDesign); ArcGIS Pro; ArcMap 10; QGIS; Python; GRASS GIS; MATLAB.

## Memberships

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American Geophysical Union  
Association for Women Geoscientists  
Geological Society of America  
National Association of Geoscience Teachers  
Society for Sedimentary Geology  
Utah Geological Association

- Updated September 8, 2026 -